

èfjç» - ç” µè`❖i¼š188-5665-3017 - é,®ç®±i¼šhs3434@foxmail.com -
ä,ªºç½ ‘ç«™i¼šhs3434.github.io - **æ±
,è❖Ææ,,❖â❖ “**i¼šAMD ROCm Python â❖Žç«`â¼â❖ ‘â • ¥ç`â, ^i¼^ROCm Radeon Cloud
æ-¹â❖ `i¼% --- ## ä,ªºç®€ä» 3 â¹´ç” Ýç%©äçjæ❖-â • ¥ç`âæ-ç»❖éâEi¼Eè ±ªâ|è½-âžç
K8s â❖Žç«`æ ^â€, \*\*â®¹â™™âæ-éf`ç½²â€❖èµ,,æº❖è`fâ|â€❖ä»»âšç¼-æž’ â€❖GPU
æž`ç❖ + æœ❖âšç¹ªâ€, æœ€èç 2 â “è ±ªâ| ROCm / K8s / Helm / FastAPI /
Prometheusi¼Eç«-ç«â®Ææ^❖ **k8s-llm-runtime**i¼^K8s ä, Šçš,, vLLM
æ`jâžçæœ❖âšç½ ‘â...³i¼%i¼E\*\*v0.1.0 taggedi¼Eâ❖`çžºâœè • ‘demo**â€, ç + Ýæ,%
Snakemake ç±» Argo Workflows çš,, ML â¹³â❖æš½è±jâ€, --- ## æ±,è❖Ææš€èf½i¼^æE%â²
—â½â❖âE¹é...❖æž’ âº❖i¼E4 â±,i¼% ### â❖Žç«`æ ,âçf - Python 3.11+i¼^asyncio, type
hints, Pydantic, httpx asynci¼% - FastAPIi¼^OpenAPI / dependency injection / lifespan¼% -
æµè` • i¼špytest, pytest-asyncio, 88% coverage ### â®¹â™™â, Žç¼-æž’ - Dockeri¼^multi-
stage æž,,âºi¼EAlpine é • œâf❖ä¼~âæ-i¼% - Kubernetesi¼^Pod / Service / Deployment /
StatefulSet / ConfigMap / RBACi¼% - Helm charti¼^values æ`jæ❖âæ-i¼Ehelm install/
upgrade/uninstalli¼Ecli lint cleani¼% - é • œâf❖â” “âº “i¼šregistry.k8s.io / docker.io /
ghcr.io / nvcr.ioi¼^â½â + ... DaoCloud é • œâf❖âš éYi¼% ### GPU / LLM ç” Ýæ€❖ -
vLLMi¼^OpenAI â...¼â®¹ APIi¼Eæž`ç❖ + æœ❖âšçj¼% - AMD ROCm âYºçççi¼^HSA / HIP
æ|,âçµi¼Eamd.com/gpu èµ,,æº❖è`fâ|i¼% - PyTorch¼^CNN æ— ¶ç©ºèš £ç ❖â™™ /
Transformeri¼% - Hugging Facei¼^transformers âº “i¼Etokenizeri¼% ###
â❖`èš ,æµæ€š ä, Žâ • ¥â... é “¾ - Prometheusi¼^prometheus_clienti¼Ehistogram /
counter / gaugei¼% - structlogi¼^JSON structured loggingi¼% - uv / ruff / mypy strict / GitHub
Actions - Linux æœ❖âšçç«`èç❖ç” i¼^systemd, Nginx, ç«`â❖E/èçç”ç®ç❖ + i¼% --- ##
éç¹ç®ç»»éâE ### k8s-llm-runtimei¼šK8s ä, Šçš,, vLLM æ`jâžçæœ❖âšç½ ‘â...³i¼^v0.1.0
taggedi¼% **é—œç~**i¼^â-¹âº” JD è❖Eè`£ 1 + 3i¼%i¼švLLM æ`jâžçéf`ç½²éœ€ GPU +
â®¹â™™ + è • -ç” ±i¼Eâçé`Yâššâºâ...±âºæ— ¶æ`jâžçç®ç❖ + âºæ❖, i¼ç” ç©¶ /
âºâçé`Yæf³ç” “ OpenAI â...¼â®¹ API æ— ¶é—æš`æš`â€, i¼^ç`æž¥â-¹âº”
JD i¼š”âšš æ`jâžçèç»f/æž`ç❖ + Workshopæ❖æ •™ç”â€âšYèf½æ¼” ç«º”i¼%
**æž ¶æž,, **i¼^â❖E chart + ä, %â±, lib + é • œâf❖ä¼~âæ-i¼%i¼š - `charts/llm-inference/
`i¼švLLM è`Yè½½½ charti¼E**GPU vendor â ±æ❖çâ»...é❖ `values.yaml` **i¼^amd.com/
gpu / nvidia.com/gpu / none ä, %æ€i¼Evalues æ~¾¼¼❖é€æ©i¼% - `charts/llm-router/
`i¼šFastAPI ç½ ‘â...³ chart - ConfigMap æ³”â...¥ initContainer âš è½½½â...â¼Y chart - RBAC
æœ€âº❖æ❖fè™i¼^Pod/Service/ConfigMap read+write + Leasei¼% - ServiceMonitor â❖`é€
%âºç” “i¼^Service 0.0.0.0:9090/metricsi¼% - `src/k8s_llm_runtime/i¼š3 â±, lib æš½è±
j - `K8sJobOperator`i¼^â½Žâ±,i¼Eé€šç” “ Job è`fâ|i¼% - `VLLMInferenceOperator`i¼^â, -
â±,i¼Ehelm CLI subprocess + GPU vendori¼% - `ModelOperator`i¼^éç`â±,i¼EOpenAI â...
¼â®¹ + è ±ªâš`éf`ç½²i¼% **4 âšš æš€æœ-âºç, **i¼š 1. **lib ä, %â±, â^ + â±, æš½è±
j**i¼^â-¹âº” JD â❖Žç«`æž ¶æž,,èf½âšçi¼% - ä, šâ±, ModelOperator â❖â...³âçf
load_model(alias) / unload_model(alias) - ä, â±, VLLMInferenceOperator â❖è£... helm
ç» + èš, - â½Žâ±, K8sJobOperator é€šç” “ K8s Job è`fâ| 2. **Helm ConfigMap æ³”â...¥
aliases**i¼^â-¹âº” JD â®¹â™™âæ- + èµ,,æº❖è`fâ|i¼% - é❖çâ...❖æ-❖æ-jæ-âçžæ`jâžçéf½
rebuild é • œâf❖i¼^GPU é • œâf❖ 15-20GB rebuild 5-10mini¼% - ConfigMap
çš’ çºšç” Ýæ • ^i¼Eé ¶é • œâf❖é ±❖âº 3. **K8s Lease â^ + â, fâ¼❖é” ❖**i¼^â-¹âº”
JD é + ç³¼èçç»’ + HAi¼% - coordination.k8s.io/v1 Lease é~²æç Router
âššâ% æœ-â¹ ¶â❖ ‘éf`ç½²â❖Eä, €æ`jâžç - é” ❖ç²’ âºæE% model
aliasi¼Eè ±ªâš`èç ±æœY + ç»çº| 4. **â, šâšçºšš Prometheus metrics**i¼^â-¹âº” JD
â¹³â❖â❖`èš ,æµæ€šš i¼% - `chat_completions_duration_seconds`i¼^histogrami¼Eâ❖«
status code æ ±ç¾¼i¼% - `http\_requests\_total`i¼^counteri¼EæE% method/route/statusi¼%

- models_loaded`i¼^gaugei¼Eä`æ` in-memory statei¼% - ServiceMonitor ä`é€%è¿
 Grafana **GPU vendor ä` ð æ` demo`\*\*i¼^ä`-1ä` JD ä` ä` + GPU é` ð ç¾äi¼%oi¼4š -
 values.yaml`gpu.vendor=amd|nvidia|none` - AMDi¼4šamd.com/gpu èµ,,æ°`è°fä°i¼4EROCm
 vLLM é • œäf` - NVIDIAi¼4šnvidia.com/gpu èµ,,æ°`è°fä°i¼4ECUDA vLLM é • œäf` -
 nonei¼4šCPU only æ` jå¼`i¼^kind cluster é`~è°äi¼4Eæœ-äœ°å¼4Eä` ‘ç` ``i¼%o` \*\*è`
 ‘`é ð` ä¿` è`-`**i¼4š - 31 commits / 90 tests pass / coverage 88% - ruff + mypy strict + helm-lint
 ä...` clean - 2 Helm chartsi¼4^llm-inference + llm-routeri¼4%o` **è, ©ä` ‘ä, Žä¿`ä°`**i¼4š -
 Service DNS ä, ä`E¹é...` bugi¼4š`--set fullnameOverride={release\_name}` ä¿`ä°` - kind
 èš,ç,¹ä½²ä` + ...é • œäf`ä` é€Yi¼4šcontainerdConfigPatches + DaoCloud mirror -
 è¿™ä°`éf½ä` +™ä...¥` docs/amd-interview-demo.md` ä¾¾ç¿`äœ°ä` ,è€f
 æœ°æ°`¥`i¼4^v1.1i¼4%oi¼4šSSE æµ`ä¼4`æŽ`ç` + / RBAC æ` ¶ ç` S ä`è` » / çœYä°Ž GPU
 é` ð ç¾äæ e2e ### äY°ä°Ž Snakemake çš,, ML
 ä¹³ä`ä` • ¥ä½²œæµ`æ; + æž¶i¼4^ç` Yä¿¿èfEæ™~i¼4Eç±» Argo Workflowsi¼4%o` -
 èfEæ™~i¼4šSnakemake rule è`æ³ • æ— æ³ • ä°žç¿`æ` jå`—és`æ,,äç»,ä`^ -
 è¾¾è°i¼4šæ` ‘ä¼4f rule è`æ³ • i¼4Eç,`æŽ`è°fä` + ...éf`æŽ`ä°Eä`S`æEç` Yæ`^
 rulei¼4Eè S,,èEfe¾ “ä...¥è¾ “ä ð°ç±»äž`ä°³ä`-ç S`-æœ`ä¼4`æ°¼æŽ`-ä...³è` i¼4šJD
 ä`S`ä` + éj¹æœ°ä™~ä`ä¹³ä`ç` ä` ‘ç»`è°E“ä” ä” æœ-éj¹ç°æ`- ML ä¹³ä`æš½è±`
 jçš,,ä` ¥ç`ä°žè • µ ### è,, ‘æœ°æŽ`ä°E CNN è S Eç` ä™~i¼4^PyTorch i¼4%oä” ä” 6 ä` ‘ä` -
 |ä¹ è • -ç¿`ä° S ä ð° - ç«`ä°ç«`-æ-¶ ç°ä` • ç S`-è S Eç` EEG ä¿¿ä` • - PyQt ä°žæ—¶ GUI +
 FastAPI æš´éœ² - è`|è S` BCI è • -ç¿`ç°Eäž` + [github.com/hs3434/learning-journey](https://
 github.com/hs3434/learning-journey/blob/main/tracks/brain-computer-interface/notes/
 resume.md) --- ## ä` ¥ä½²œç»`äž` + ### ä, Šæµ • ä°é ð Eä¿¿ä¿¿ä`-ç S` ‘æš€æœ°%é™ä...
 -ä` , i½œ ç` Yç°%©ä¿¿ä`-ä` ¥ç`ä, ^ i½œ 2025.04 - è ð³ä`S` - **Docker
 éf`ç½²\*\*i¼4šä, šäš¿ä»Eç` Docker äE-i¼4^äY°ä°Ž snakemake + è ð°ç` æ; + æž¶i¼4%o` -
 **æœ°ä`š¿ç«`-è¿¿ç»`\*\*i¼4šLinux æœ°ä`š¿ä™~æ-¥ä, ç»`æš¿i¼4Enginx
 ä`ä`ä` ‘ä»Eç` + i¼4Esystemd æœ°ä`š¿ç°¿ç` + - **ä°` ä¹³ä`ä` • ¥ä...
 • ä¼4Eä` \*\*i¼4šä, šäš¿è ð ä`S`äE-æµ`ç`+ç°—æ³ • ä` ¥ä... • ###
 ä, Šæµ • æ` S æ` “ç” Yç°%©äE»ä¿¿ S` ‘æš€æœ°%é™ä...-ä` , i½œ
 ç” Yç°%©ä¿¿ä`-ç` ä` ‘ä` ¥ç`ä, ^ i½œ 2022.08 - 2024.07 - \*\*ä°` ä¹³ä`ç°` Yä¿¿ä` • ¥ä...
 • ä¼4Eä` \*\*i¼4šR / Python / Linux / Docker / ä°` äžYç” Y`- \*\*ä, šäš¿è ð ä`S`äE- \*\*i¼4šäE»ä` -
 |è½²-ä½² • ç»,ä, šäš¿æµ`ç«`è ð ä`S`äE-i¼4Eæ` jå`—äE-äEä`ä`-ä°ç”` -
 ä¹³ä`äœ°ä`€i¼4š[cloud.oebiotech.com](https://cloud.oebiotech.com/#/home) ###
 ä, Šæµ • æ` S æ` “ç” Yç°%©äE»ä¿¿æœ°%é™ä...-ä` , i½œ ç” Yç°%©ä¿¿ä`-ä°žä¹ ç” Y`i½œ
 2021.10 - 2022.01 - ä` • ç»` + èfžè½²-ä½² • ç»,i¼4šè
 ‘`æž S äEä, è S,,ä` + æžäEä` ä`žäEä, äE S äE-ä` + æž` -
 ä, »è¿`æš€æœ- i¼4šä` • ç»` + èfžè½²-ä½² • ç»,äEä`RäEä`PythonäEä`Linux --- ##
 æ • ™è,²èfEæ™~` \*\*è¥¿äE—ä` + œæž-ç S` ‘æš€ä° S ä¿¹ i½œ æ°`ç°%©ç S` ‘ä` -
 |ä, Žæš€æœ-ä, “ä, š i½œ æœ-ç S` ‘i½œ 2018.09 - 2022.07 i½œ GPAi¼4š3.35 - æ` • ä, šè°æ-
 ð i¼4šäEšäY°ä°Žé€»è¾ “æ-è” ,äžä½²` ç¿`ä°šä°`é°¿ä” ä¼4`ç», ä...³æE S çš¶ ä`šäS—
 ä” æE S çš¶ æE ð æ ð äE- ä° S ä`éj¹ç°i¼4šæ¹ç°èèEææžç S` ä`¹ç`«èš±èœè “¿äS—
 é “œæ±¿æY “çš,,ä½²±ä` “äi¼4^ä¿,ä, Žä½²œèE...i¼4%o` --- ## ä, ä°°ä¼4^äš¿ - **6 ä` ‘è ð ä¿ K8s
 ä°žç«`-æ`**i¼4šä`žé¶ä° v0.1.0 tagged éj¹ç°i¼4Eç«`-ç«`ä°Eæ`ä°Eä` Helm chart + Python
 lib - **ç¿`äœ° demo èf½äš`**i¼4šä`-è ð ä, |ç” µè,, ‘äœ` kind cluster ä, šè • ‘éšä°Eæ •`
 LLM æž`ç` + æµ`ç»`i¼4^éf`ç½²ä` +` è • -ç” ± ä` +` chat completions ä` +` metricsi¼4%o` -
 **ä¿¹ èf½äš`ä¼4°\*\*i¼4šä`ä¹`ä` + ...ä»žç” Yç°%©ä¿¿ä`-è • -ç • Eä° K8s
 ä°žç«`-i¼4Eæš€æœ-çfæf...è~ - \*\*3ä¹` Python ä` ¥ç`äE-
 ç»`è°Eæ\*\*i¼4šä»žç S` ‘ç” è,,šæœ-ä°ä°` ä¹³ä`ç°`ä` ¥ä... • çš,,ä°Eæ •`è • -ä¾¾,,